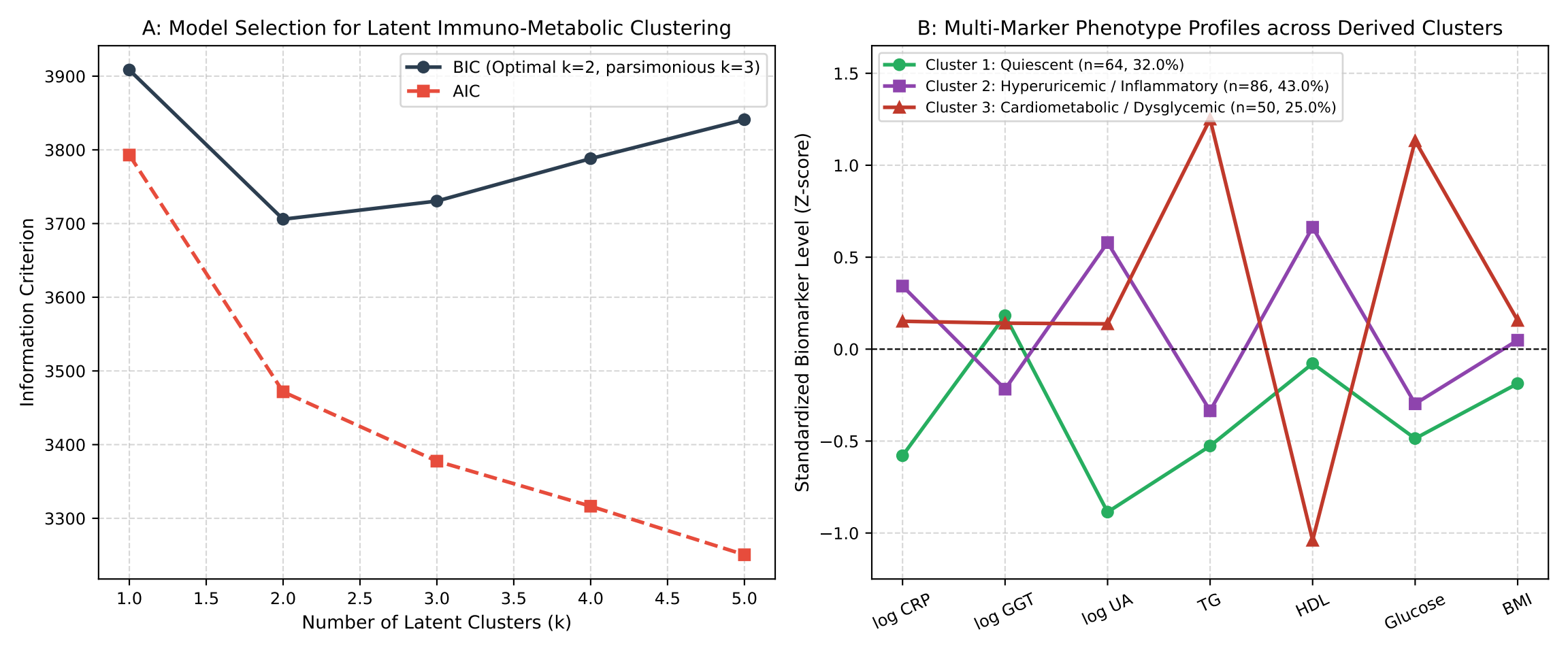
A: Model Selection for Latent Immuno-Metabolic Clustering
BIC (Optimal k=2, parsimonious k=3)
AIC
Information Criterion
Number of Latent Clusters (k)
3900
3800
3700
3600
3500
3400
3300
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
B: Multi-Marker Phenotype Profiles across Derived Clusters
Cluster 1: Quiescent (n=64, 32.0%)
Cluster 2: Hyperuricemic / Inflammatory (n=86, 43.0%)
Cluster 3: Cardiometabolic / Dysglycemic (n=50, 25.0%)
Standardized Biomarker Level (Z-score)
1.5
1.0
0.5
0.0
−0.5
−1.0
log CRP
log GGT
log UA
TG
HDL
Glucose
BMI